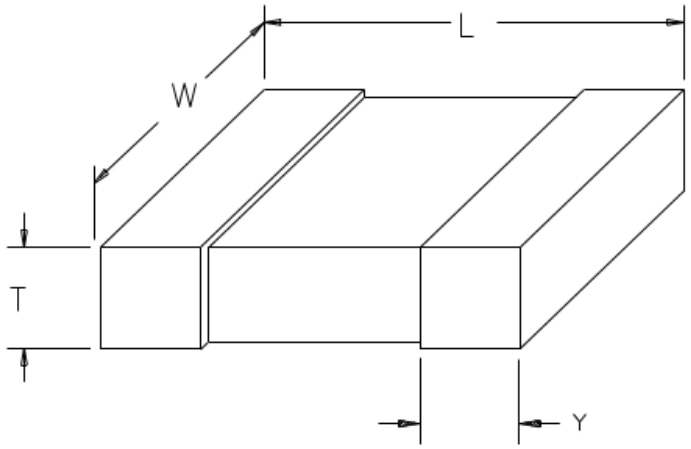
INCH-POUND
MIL-PRF-32535/7A
5 September 2023
SUPERSEDING
MIL-PRF-32535/7
w/AMENDMENT 2
4 June 2018

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, FIXED, CERAMIC DIELECTRIC (TEMPERATURE STABLE AND GENERAL PURPOSE), EXTENDED RANGE, HIGH RELIABILITY AND STANDARD RELIABILITY, SIZE 1812

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32535.



Dimensions			
L ± .012	W ± .012	T Max.	Y ± .018
.178	.126	.110	.024

inches	mm
.008	0.20
.012	0.30
.014	0.36
.024	0.61
.110	2.79
.126	3.20
.178	4.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for information only.
3. Dimensions and tolerances are for terminated chips.
4. For solder termination finishes, add .025 inch (0.64 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerances. The increase in dimension applies to the solder coating thickness only.

FIGURE 1. Size 1812 capacitors.

AMSC N/A
FSC 5910
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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Voltage-temperature limit or temperature characteristic (VTL/TC): BP, C0G, and X7R as specified in MIL-PRF-32535.

Rated voltage (V_{dc}): V = 4; W = 6.3; X = 10; Y = 16; Z = 25; A = 50; B = 100; C = 200. See table I for maximum rated voltage available for each capacitance value.

Operating temperature range: -55°C to +125°C.

Termination finish: D, G, H, M, R, V, and Z as specified in MIL-PRF-32535.

Electrode: P and B as specified in MIL-PRF-32535.

Product level designator: Standard reliability – M and high reliability - T.

Marking: In accordance with MIL-PRF-32535.

TABLE I. Size 1812 capacitor characteristics.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	VTL/TC	Rated voltage 2/ (V_{dc})	Electrode material
M3253507 - - - 102 - - - -	1,000	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 122 - - - -	1,200	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 152 - - - -	1,500	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 182 - - - -	1,800	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 222 - - - -	2,200	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 272 - - - -	2,700	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 332 - - - -	3,300	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 392 - - - -	3,900	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 472 - - - -	4,700	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 562 - - - -	5,600	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 682 - - - -	6,800	F, G, J, K	BP, C0G	200	P, B
M3253507 - - - 822 - - - -	8,200	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 103 - - - -	10,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 123 - - - -	12,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 153 - - - -	15,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 183 - - - -	18,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 223 - - - -	22,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 273 - - - -	27,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 333 - - - -	33,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 393 - - - -	39,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 473 - - - -	47,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 563 - - - -	56,000	F, G, J, K	BP, C0G	100	P, B
M3253507 - - - 683 - - - -	68,000	F, G, J, K	BP, C0G	50	P, B
M3253507 - - - 823 - - - -	82,000	F, G, J, K	BP, C0G	25	P, B
M3253507 - - - 104 - - - -	100,000	F, G, J, K	BP, C0G	25	P, B
M3253507E2 - 104 - - - -	100,000	J, K, M	X7R	100	P, B
M3253507E2 - 154 - - - -	150,000	J, K, M	X7R	100	P, B

See footnotes at end of table.

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TABLE I. Size 1812 capacitor characteristics - Continued.

PIN 1/	Capacitance (pF)	Capacitance tolerance	VTL/TC	Rated voltage 2/ (V_{dc})	Electrode material
M3253507E2 - 184 - - - -	180,000	J, K, M	X7R	100	P, B
M3253507E2 - 224 - - - -	220,000	J, K, M	X7R	100	P, B
M3253507E2 - 274 - - - -	270,000	J, K, M	X7R	100	P, B
M3253507E2 - 334 - - - -	330,000	J, K, M	X7R	100	P, B
M3253507E2 - 394 - - - -	390,000	J, K, M	X7R	100	P, B
M3253507E2 - 474 - - - -	470,000	J, K, M	X7R	100	P, B
M3253507E2 - 564 - - - -	560,000	J, K, M	X7R	100	P, B
M3253507E2 - 684 - - - -	680,000	J, K, M	X7R	100	P, B
M3253507E2 - 824 - - - -	820,000	J, K, M	X7R	100	P, B
M3253507E2 - 105 - - - -	1,000,000	J, K, M	X7R	100	P, B
M3253507E2 - 125 - - - -	1,200,000	J, K, M	X7R	100	P, B
M3253507E2 - 155 - - - -	1,500,000	J, K, M	X7R	100	P, B
M3253507E2 - 185 - - - -	1,800,000	J, K, M	X7R	100	P, B
M3253507E2 - 225 - - - -	2,200,000	J, K, M	X7R	100	P, B
M3253507E2 - 275 - - - -	2,700,000	J, K, M	X7R	50	P, B
M3253507E2 - 335 - - - -	3,300,000	J, K, M	X7R	50	P, B
M3253507E2 - 395 - - - -	3,900,000	J, K, M	X7R	50	P, B
M3253507E2 - 475 - - - -	4,700,000	J, K, M	X7R	50	P, B
M3253507E2 - 565 - - - -	5,600,000	J, K, M	X7R	25	P, B
M3253507E2 - 685 - - - -	6,800,000	J, K, M	X7R	25	P, B
M3253507E2 - 825 - - - -	8,200,000	J, K, M	X7R	25	P, B

1/ The complete PIN shall include additional symbols to indicate VTL/TC (where applicable), voltage, capacitance tolerance, termination finish, product level, and electrode material.
2/ This is the maximum rated voltage available. All lower voltage ratings are also available.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2023-025)

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19
Other – MDA, NA

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